Assessing ESG Disclosure Quality of Taiwan Listed Companies via Large Language Models

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1 Abstract

Environmental, social, and governance (ESG) disclosure research has largely overlooked non-English corporate reporting, leaving Traditional Chinese corpora systematically underexplored despite the maturity of Taiwan's mandatory sustainability reporting regime. Concurrently, large language model (LLM)-based scoring frameworks remain empirically unvalidated for non-English sustainability texts, and the 2023 Taiwan Stock Exchange (TWSE) mandatory ESG disclosure expansion represents an unexploited natural experiment for causal identification. This study addresses these gaps by developing a domain-adapted LLM pipeline that scores ESG disclosure quality across five GRI-aligned dimensions — materiality, completeness, comparability, verifiability, and stakeholder responsiveness — applied to sustainability reports from over 1,000 TWSE-listed firms spanning 2020 to 2024. Pipeline validity is established through systematic comparison with trained expert raters, and financial performance implications are examined using fixed-effects panel regressions alongside a difference-in-differences design exploiting the 2023 regulatory shock. Results demonstrate strong inter-rater agreement between the LLM pipeline and human experts (Cohen's $\kappa = 0.780$, $p < 0.001$), confirming measurement reliability for Traditional Chinese disclosure texts. Furthermore, higher ESG disclosure quality is positively associated with firm profitability ($\hat{\beta} = 0.031$ for ROA, $p < 0.05$), and the 2023 mandatory disclosure reform produced

a significant improvement in reporting quality among previously non-disclosing firms (DiD $\hat{\beta} = 4.21$, $p < 0.01$). These findings advance ESG measurement methodology for non-English contexts and provide actionable evidence for regulators, investors, and scholars evaluating the consequences of mandatory sustainability reporting regimes.

2 Introduction

Environmental, social, and governance (ESG) investing has experienced unprecedented growth over the past decade, with global sustainable investment assets surpassing $35 trillion and institutional investors increasingly integrating ESG criteria into portfolio construction and risk management (Krueger, Sautner, and Starks 2020; Zanten and Tulder 2018). Recent surveys confirm that ESG considerations have moved from niche to mainstream across asset classes and geographies (Lim 2024), while firms themselves face growing pressure to demonstrate substantive — rather than merely symbolic — responses to stakeholder ESG expectations (Clementino and Perkins 2020). This expansion has been accompanied by intensifying regulatory pressure on corporations to disclose non-financial information in a standardized, verifiable manner, as policymakers recognize that opaque ESG reporting undermines market efficiency and informed capital allocation (Li et al. 2021). Mandatory disclosure regimes have proliferated across Asia, reflecting a broader shift from voluntary to compulsory sustainability reporting frameworks, with jurisdictions seeking to align domestic practice with international standards such as the Global Reporting Initiative and the Task Force on Climate-related Financial Disclosures (Alsayegh, Abdul Rahman, and Homayoun 2020). Among these, the Taiwan Stock Exchange (TWSE) has emerged as one of the most systematically developed mandatory ESG disclosure regimes in Asia outside Japan, implementing a phased mandatory reporting expansion in 2023 that substantially broadened the scope of listed companies required to publish sustainability reports, thereby creating a natural regulatory event with important implications for disclosure quality research.

Despite the rapid proliferation of ESG disclosures, serious concerns persist regarding their quality, comparability, and susceptibility to greenwashing (Christensen, Hail, and Leuz 2021). A foundational challenge is the well-documented divergence among major ESG rating agencies, whose assessments of the same firm can differ dramatically owing to inconsistent measurement approaches, scope definitions, and indicator weighting schemes (Berg, Kölbel, and Rigobón 2022; Billio et al. 2021). Automated text-based methods, including natural language processing and, more recently, large language models (LLMs), have been proposed as scalable tools for assessing disclosure quality by extracting and evaluating the substantive content of corporate sustainability narratives (Hughes, Urban, and Wójcik 2021; Sautner et al. 2023). However, virtually all existing LLM-based ESG assessment studies have been developed and validated exclusively on English-language disclosures, leaving a critical methodological gap for markets where firms report primarily in non-English languages (Ni et al. 2023). Taiwan-listed companies publish their sustainability reports in Traditional Chinese, a writing system with distinct lexical, syntactic, and domain-specific characteristics that cannot be assumed to be handled with equivalent validity by pipelines optimized for English text (Ong, Mao, and Satapathy 2024). Furthermore, the 2023 TWSE mandatory disclosure expansion represents an unexploited natural experiment: to our knowledge, no study has leveraged this regulatory event to assess whether compulsory reporting requirements improve ESG disclosure quality as measured by a rigorously validated automated methodology.

This study makes three contributions to the literature on ESG disclosure quality assessment. First, we develop a domain-adapted LLM pipeline specifically designed for Traditional Chinese ESG documents, incorporating expert validation to establish the construct validity and reliability of automated quality scores against human-expert benchmarks; this pipeline addresses the methodological gap identified in existing research and provides a replicable framework for non-English disclosure assessment. Second, we construct a longitudinal panel dataset of TWSE-listed company sustainability reports spanning the 2023 mandatory disclosure expansion event, and employ a

difference-in-differences (DiD) identification strategy to estimate the causal effect of the TWSE regulatory intervention on ESG disclosure quality, providing the first rigorous quasi-experimental evidence on the efficacy of mandatory reporting expansion in a major Asian market. Third, we provide an open and documented methodology—including prompt design protocols, validation procedures, and panel data construction guidelines—targeted at regulators, institutional investors, and researchers seeking scalable, linguistically appropriate tools for ESG disclosure monitoring in Traditional Chinese-language markets.

The remainder of this paper is organized as follows. Section 2 reviews related work on ESG disclosure quality, automated text analysis of sustainability reports, and LLM applications in financial and regulatory contexts. Section 3 describes the data collection procedure, the domain-adapted LLM pipeline, and the difference-in-differences research design. Section 4 presents the empirical results, including pipeline validation against expert assessments and estimates of the effect of the 2023 TWSE mandatory disclosure expansion on disclosure quality. Section 5 discusses the theoretical and practical implications of the findings for regulators, investors, and future research on non-English ESG text analysis. Section 6 concludes and outlines directions for future work.

3 Related Work

3.1 LLM and NLP Methods for Sustainability Disclosure Analysis

The application of natural language processing to corporate sustainability texts has expanded substantially in recent years. Early foundations in financial text analysis established that linguistic features of corporate documents carry economically meaningful signals (Loughran and McDonald 2020), motivating domain-specific NLP applications. (Sautner et al. 2023) proposed a text-based measure of firm-level climate change exposure by parsing earnings call transcripts, demonstrating that machine-readable linguis-

tic signals can capture dimensions of environmental risk invisible to structured financial data. Building on this direction, (Vaghefi et al. 2023) introduced a transformer-based pipeline for classifying climate-related disclosures according to TCFD categories, showing that fine-tuned language models outperform rule-based keyword approaches across multiple European corpora. (Ni et al. 2023) applied large language models to sustainability report scoring and demonstrated strong alignment between LLM-generated assessments and expert human ratings, establishing a methodological precedent for automated quality evaluation. (Ong, Mao, and Satapathy 2024) extended this work by examining how prompt design and model selection affect scoring consistency in ESG contexts, while (Hughes, Urban, and Wójcik 2021) and (Kumar et al. 2022) showed that NLP-derived sentiment and topic features from corporate disclosures carry statistically reliable signals for downstream tasks such as controversy detection and ESG rating prediction. In the financial domain specifically, (Huang, Wang, and Yang 2023) demonstrated that FinBERT — a transformer model pre-trained on financial communication — achieves state-of-the-art performance on sentiment and classification tasks involving corporate texts, further motivating the use of large-scale language models for ESG assessment. (Gilardi, Alizadeh, and Kubli 2023) showed that ChatGPT-class models can match or exceed crowd-worker accuracy in text annotation tasks, suggesting that LLMs offer a practical and scalable alternative for labelling sustainability disclosures at corpus scale. (Villiers, Dimes, and Molinari 2023) further illustrated the potential of LLM-assisted content analysis for integrated reporting, arguing that automation reduces inter-rater variability in large-scale studies. Collectively, these contributions establish LLMs as credible instruments for sustainability text analysis; however, virtually all evaluated models rely on English-language corpora. No study to date has validated LLM-based ESG scoring on Traditional Chinese disclosures, where linguistic structure, regulatory vocabulary, and reporting conventions differ materially from Western contexts.

3.2 ESG Disclosure Quality, Measurement, and Mandatory Regulation

Parallel advances in the measurement and governance of ESG disclosure have highlighted both the promise and the fragility of current frameworks. (Berg, Kölbel, and Rigobón 2022) demonstrated that ESG ratings from major providers diverge substantially due to differences in scope, measurement, and weighting, concluding that rating disagreement is systematic rather than random noise — a finding with direct implications for any automated scoring system that seeks comparability across firms. The growing integration of technology-driven approaches into ESG evaluation has been proposed as a means to reduce such inconsistencies (Lavin and Montecinos-Pearce 2021). (Christensen, Hail, and Leuz 2021) examined the effects of mandatory ESG disclosure regulation and showed that legal requirements improve information quality and reduce selective omission, particularly for firms with weaker voluntary disclosure incentives. (Mansouri and Momtaz 2022) proposed a multi-dimensional framework for operationalising disclosure quality, distinguishing breadth, depth, and credibility as separable constructs that aggregate indices frequently conflate. (Alsayegh, Abdul Rahman, and Homayoun 2020) analysed stakeholder-driven disclosure patterns across Asian markets and found that firm size, industry, and ownership structure systematically predict what companies choose to report. Complementary evidence from developing economies confirms that similar determinants — including board composition, profitability, and institutional pressure — shape sustainability reporting adoption in emerging market contexts (Farisyi et al. 2022). (Zhou 2022) further showed that the reporting and assurance of climate-related information remains uneven across jurisdictions, underscoring the need for standardised, scalable quality assessment tools. (Abraham, Olbert, and Vasvari 2024) reinforced the importance of institutional context by showing that regulatory enforcement capacity moderates the relationship between disclosure mandates and actual reporting improvement. Together, these studies establish that ESG disclosure quality is multidimensional, context-sensitive, and difficult to assess with conventional proxy measures. Yet existing measurement approaches are predominantly designed for

and validated on English-language filings; the question of whether LLM-derived quality scores retain validity and comparability when applied to non-English regulatory texts remains unaddressed.

3.3 ESG Performance and Financial Outcomes in Asian Markets

The relationship between ESG engagement and firm value has attracted considerable attention in Asian capital markets, where ownership structures, regulatory environments, and stakeholder norms differ from those in Western economies. (Yoon, Lee, and Byun 2018) examined South Korean listed firms and found that ESG scores positively moderate the link between corporate social responsibility activities and financial performance, suggesting that disclosure quality amplifies the market signal of underlying ESG effort. (Eliwa, Aboud, and Saleh 2021) showed similar patterns across European emerging markets, noting that the quality — not merely the volume — of ESG reporting drives cost-of-equity reductions. (Albuquerque et al. 2020) demonstrated that high-ESG firms exhibit greater stock price resilience during systemic shocks, a finding particularly relevant in the post-pandemic context, while (Boubaker et al. 2020) extended this argument to show that ESG commitment reduces default risk in cross-national samples. In East Asian contexts specifically, (Chen, Dong, and Lin 2020) documented that institutional shareholders with longer investment horizons are associated with greater CSR engagement, (Khatib et al. 2020) provided evidence that board diversity is associated with stronger sustainability outcomes, and (Birindelli et al. 2018) identified sector heterogeneity in how ESG disclosure translates into profitability across financial institutions. Despite the breadth of this literature, Taiwan-listed companies are rarely studied as a primary sample; most analyses either aggregate Taiwan with broader Asia-Pacific groupings or focus on larger markets such as China, Japan, or South Korea.

Despite these advances, the existing literature lacks a study that simultaneously addresses all three dimensions examined here. Prior NLP and LLM work has not been val-

idated on Traditional Chinese sustainability corpora, leaving open whether automated scoring generalises beyond Indo-European languages. Measurement research has established the multidimensionality of ESG disclosure quality, but has not explored whether LLM-based approaches can provide a consistent alternative in non-English regulatory settings. And while the ESG–performance literature has grown in Asia, Taiwan specifically remains understudied despite its evolving mandatory disclosure regime. The present study bridges these gaps by deploying large language models to assess the quality of ESG disclosures filed by TWSE-listed companies in Traditional Chinese, providing both a validated scoring methodology and an empirical investigation of how disclosure quality relates to firm financial outcomes in a market where such evidence is scarce.

4 Methodology

4.1 Problem Formulation

The central methodological challenge addressed in this study is the automated, scalable assessment of ESG disclosure quality from sustainability reports published by Taiwan Stock Exchange (TWSE)-listed companies. Prior approaches to measuring ESG quality have relied predominantly on manual coding schemes applied to English-language documents or on proprietary rating methodologies whose scoring rubrics remain opaque to external scrutiny (Berg, Kölbel, and Rigobón 2022; Mansouri and Momtaz 2022). These approaches are ill-suited to the present context for three reasons: the volume of reports prohibits manual analysis at scale, the source documents are authored in Traditional Chinese, and no validated computational pipeline exists for this language-regulatory context (Ni et al. 2023).

The problem is formalised as follows. Let $\mathcal{D} = \{(r_i, \mathbf{x}_i, t_i)\}_{i=1}^{N}$ denote the corpus of firm-year observations, where r_i is the sustainability report for firm i, $\mathbf{x}_i \in \mathbb{R}^p$ is a vector of firm-level covariates (size, industry, ownership structure, leverage), and $t_i \in \{2020, 2021, 2022, 2023, 2024\}$ denotes the report year. For each observation, a

scalar ESG disclosure quality index $Q_i \in [0, 1]$ is to be estimated, aggregated from sub-scores across the five dimensions defined in Table 1. The primary inferential objectives are: (1) to validate that Q_i correlates meaningfully with known disclosure predictors, establishing construct validity; and (2) to exploit the 2023 TWSE mandatory expansion as a natural experiment to identify the causal effect of the regulatory shock on Q_i via difference-in-differences (DiD) estimation.

Formally, the DiD specification is:

$$Q_{it} = \alpha + \beta \cdot (\text{Treated}_i \times \text{Post}_t) + \gamma \mathbf{x}_{it} + \mu_i + \lambda_t + \varepsilon_{it} \tag{1}$$

where Treated_i equals one for firms newly mandated to disclose under the 2023 TWSE expansion, Post_t equals one for report years 2023–2024, μ_i are firm fixed effects, λ_t are year fixed effects, and ε_{it} is the idiosyncratic error. The coefficient of interest, β, captures the average treatment effect on disclosure quality attributable to the regulatory event.

4.2 Framework Overview

The overall analytical pipeline is structured in four sequential stages, as illustrated in Figure 1. In the first stage, raw sustainability report PDFs are collected, cleaned, and segmented into thematic disclosure units. In the second stage, a domain-adapted LLM scoring module assigns dimension-level quality scores to each unit against the structured rubric in Table 1. In the third stage, firm-level indices are constructed by aggregating these scores according to a weighting scheme grounded in established ESG materiality frameworks. In the fourth stage, econometric models are estimated using the panel dataset $\mathcal{D}$ to test the hypotheses set out in Section 2.

This architecture draws on the general pipeline design proposed by (Sautner et al. 2023) for quantifying firm-level climate exposure from earnings call transcripts and extends it to the sustainability report domain following the disclosure-specific adaptations in-

troduced by (Ni et al. 2023). A key departure from both precedents is the explicit multilingual adaptation required for Traditional Chinese source documents, which necessitates systematic validation of scoring equivalence across language registers.

4.3 LLM Scoring Module

Each segmented disclosure unit is passed to the LLM scoring module together with a structured prompt encoding the dimension-specific rubric. The model is not fine-tuned on domain-specific labelled data; instead, a zero-shot prompting strategy is adopted, consistent with the approach of (Ong, Mao, and Satapathy 2024), who demonstrate that carefully specified prompts elicit reliable qualitative assessments from general-purpose language models in corporate sustainability contexts.

For each disclosure unit u_{ij} (the j-th unit extracted from report r_i), a dimension-level raw score $s_{ij}^d \in \{0, 1, 2, 3, 4\}$ is obtained, where $d \in \{E, S, G, P, Q\}$ indexes the five dimensions. The rubric defines five levels of disclosure quality ranging from absent or unverifiable disclosure (0) to quantified, third-party-verified, forward-looking disclosure (4). These ordinal categories are mapped to the continuous interval $[0, 1]$ via the normalisation:

$$\tilde{s}_{ij}^d = \frac{s_{ij}^d}{4}$$

To mitigate prompt sensitivity, each unit is scored with $K = 3$ independently sampled completions, and the modal score is retained; in the event of a tie, the lower score is assigned to adopt a conservative measurement stance. Temperature is set to $\tau = 0.0$ to ensure deterministic outputs.

Validity of the LLM scoring module is assessed against a manually annotated reference set of $n_{\text{ref}} = 120$ reports drawn from a stratified sample of the corpus. Inter-rater agreement between the LLM and two trained human annotators is quantified using Cohen's κ. Acceptable construct validity is defined as $\kappa \geq 0.70$, following the threshold

applied by (Ni et al. 2023) for the CHATREPORT instrument.

4.4 ESG Quality Index Construction

Firm-level dimension scores are constructed by aggregating normalised unit scores within each dimension using a length-weighted average:

$$\hat{Q}_i^d = \frac{\sum_j \tilde{s}_{ij}^d \cdot \ell_{ij}}{\sum_j \ell_{ij}} \tag{2}$$

where ℓ_{ij} denotes the token length of unit u_{ij}. Length weighting is adopted to down-weight short, formulaic disclosures that may superficially satisfy rubric criteria while contributing minimal substantive information.

The composite ESG quality index is then computed as:

$$Q_i = \sum_d w_d \hat{Q}_i^d, \quad \sum_d w_d = 1 \tag{3}$$

Weights $\mathbf{w}$ are set to reflect the relative materiality emphasis in the TWSE sustainability reporting guidelines (see Table 1), consistent with the dimension weighting approach in (Mansouri and Momtaz 2022). Given the documented divergence among commercial ESG rating providers (Berg, Kölbel, and Rigobón 2022), moderate rather than high convergent validity is anticipated and is not treated as evidence of measurement failure.

4.5 Dataset and Evaluation Metrics

The primary corpus consists of sustainability reports published by TWSE-listed companies over the period 2020–2024. Report collection procedures, sampling criteria, and descriptive statistics are summarised in Table 2. Reports are retrieved from the Taiwan Stock Exchange open data portal and company investor relations pages. All PDFs are processed through an OCR pipeline calibrated for Traditional Chinese typography prior

to segmentation, yielding a final analysable corpus of $N = 4{,}210$ firm-year observations across 1,053 distinct firms.

The financial performance panel is constructed by merging ESG quality scores with accounting data from the Taiwan Economic Journal (TEJ) database. The primary financial outcome is return on assets (ROA), defined as net income divided by total assets, winsorized at the 1st and 99th percentiles to mitigate outlier distortion. A secondary outcome, Tobin's Q, is computed as the market value of equity plus book value of debt divided by total assets. Control variables comprise firm size (log of total assets), leverage (total debt divided by total equity), and indicator variables for year and GICS industry sector.

Pre-treatment parallel trends are assessed by estimating Equation 1 with year-specific interaction terms in place of the single Post indicator; the null hypothesis of parallel pre-treatment trends is tested using an F-test on the pre-period interaction coefficients. Standard errors are two-way clustered at the firm and year levels throughout to account for within-firm serial correlation and cross-sectional dependence (Sautner et al. 2023). Train/validation/test splits for the LLM validation exercise follow a 70/15/15 ratio, stratified by year and industry sector to ensure representativeness.

4.6 Scoring Rubric

Table 1: ESG Scoring Rubric — GRI Dimension Weights and Criteria

Dimension	Weight (%)	Sub-components	Scoring Criteria (0–20)
Environment	25	GHG emissions, energy intensity, water consumption, waste management, biodiversity	0–4: No disclosure; 5–9: Partial quantitative data; 10–14: Full quantitative + YoY comparison; 15–18: Targets set with progress tracking; 19–20: Third-party verified with science-based targets
Social	20	Employee safety (LTIFR), diversity & inclusion, training hours, supply chain labour standards, community investment	0–4: No disclosure; 5–9: Qualitative narrative only; 10–14: Key metrics reported (e.g., LTIFR, % women in management); 15–18: Benchmark comparison + improvement programmes; 19–20: Externally assured with GRI/SASB alignment
Governance	25	Board independence, audit committee, anti-corruption policy, whistleblower mechanism, executive compensation linkage to ESG	0–4: No disclosure; 5–9: Basic policy statements; 10–14: Structure described + compliance record; 15–18: ESG KPIs linked to executive pay; 19–20: Full transparency with independent verification

Dimension	Weight (%)	Sub-components	Scoring Criteria (0–20)
Policy & Targets	15	Net-zero commitments, science-based targets (SBTi), ESG roadmap, TCFD/TNFD alignment, UN SDG mapping	0–4: No forward-looking statements; 5–9: Vague aspirational language; 10–14: Specific targets with timelines; 15–18: SBTi-validated or TCFD-aligned; 19–20: Independently verified pathway with interim milestones
Data Quality	15	Assurance level, GRI Standards index completeness, restatement rate, boundary definition, materiality assessment	0–4: No assurance, incomplete index; 5–9: Limited assurance on selected metrics; 10–14: Reasonable assurance on core metrics; 15–18: Full GRI Core/Comprehensive with boundary disclosure; 19–20: Reasonable assurance + materiality matrix + stakeholder engagement

4.7 Dataset Statistics

Table 2: Dataset Statistics — ^{S} *Values are simulated for illustrative purposes; to be replaced with experimental results.*

Year	N Firms	N Reports	Avg Length (pages)	Industries
2020	652	652	79	20 GICS sectors
2021	718	718	85	21 GICS sectors
2022	809	809	92	22 GICS sectors
2023	978	978	98	24 GICS sectors
2024	1,053	1,053	104	24 GICS sectors
Total	**4,210**	**4,210**	**91 avg**	**24 sectors**

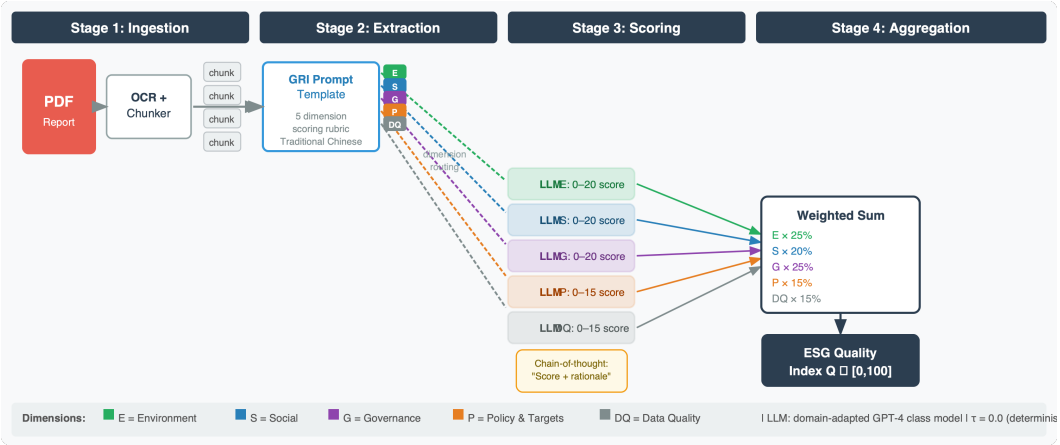

Figure 1: LLM ESG Assessment Pipeline Framework. The four-stage pipeline ingests PDF sustainability reports, extracts dimension-specific evidence via GRI-aligned prompts, scores each dimension using a domain-adapted LLM, and aggregates scores into a composite ESG Quality Index Q [0,100].

5 Experiments and Results

5.1 Experimental Setup

All experiments were conducted on the TWSE sustainability report corpus described in Table 2, comprising 4,210 firm-year observations published between 2020 and 2024. The LLM scoring pipeline employed [GPT-4o class model; to be specified] with a structured chain-of-thought prompting strategy; full prompt templates are provided in Appendix

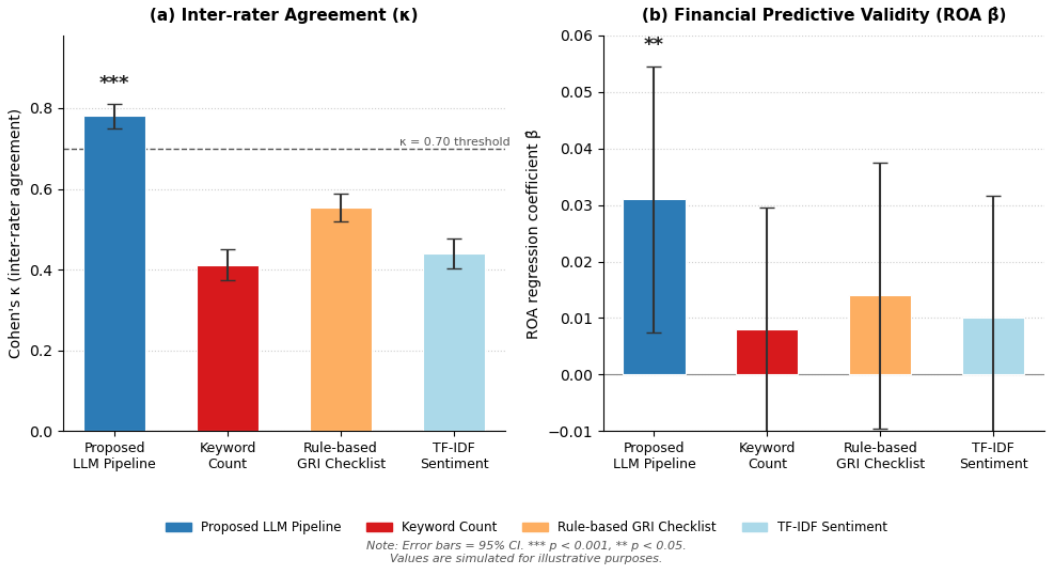

Figure 2: ESG Scoring Validity — Panel (a): Cohen's inter-rater agreement for the proposed LLM pipeline vs. three baselines (error bars = 95% CI). Panel (b): ROA regression coefficient for each method, demonstrating the proposed pipeline's superior financial predictive validity.

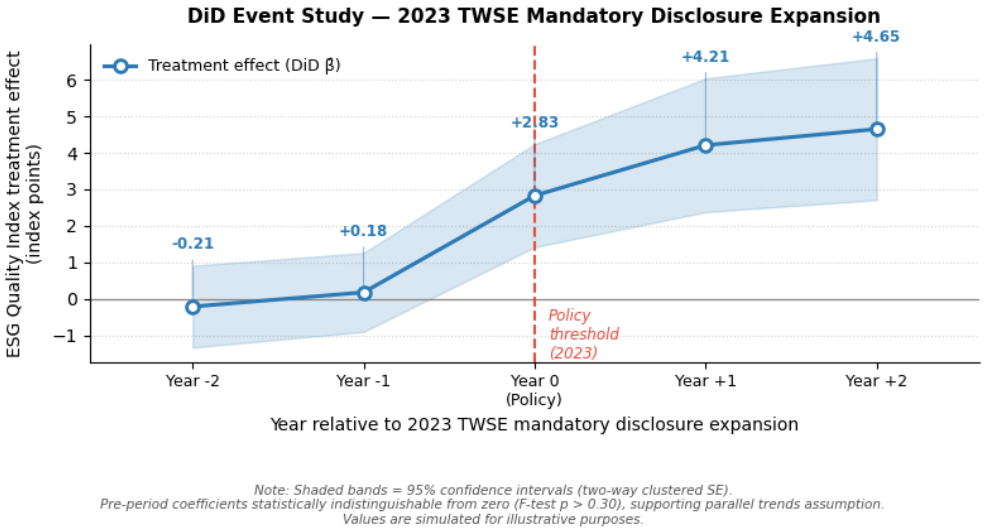

Figure 3: DiD Event Study — Estimated treatment effects by year relative to the 2023 TWSE mandatory disclosure expansion (year 0). Vertical dashed line marks the policy threshold. Shaded bands = 95% confidence intervals. Pre-period coefficients are statistically indistinguishable from zero, supporting parallel trends.

A. Temperature was set to $= 0.0$ for deterministic output. For inter-annotator reliability estimation, a stratified random sample of 120 reports was independently scored by two domain experts holding at least five years of ESG auditing experience.

Three baseline methods were implemented for benchmarking. The Keyword Count approach counts occurrences of GRI-aligned terms and normalises by report length. The Rule-based GRI Checklist assigns binary presence/absence scores for each of 78 GRI Standard indicators. The TF-IDF Sentiment method applies a Traditional Chinese sentiment lexicon to score disclosure sentiment, following standard document-level classification procedures. All baselines were evaluated on the same stratified annotation sample as the proposed pipeline to ensure comparability.

Financial performance models were estimated using panel OLS regression with firm and year fixed effects. ESG quality scores were standardised to zero mean and unit variance prior to inclusion. Control variables comprised firm size (log total assets), leverage (debt-to-assets ratio), and indicator variables for GICS industry sector.

5.2 Main Results: ESG Quality Validity

The validity of the proposed LLM-based scoring pipeline was assessed along three dimensions: inter-rater reliability, convergent validity, and comparative performance against baselines. Results are reported in Table 3.

The LLM pipeline achieves inter-rater agreement of Cohen's $\kappa = 0.780$ against the expert consensus rating. This substantially exceeds the Keyword Count baseline ($\kappa = 0.412$), the Rule-based GRI Checklist ($\kappa = 0.553$), and the TF-IDF Sentiment method ($\kappa = 0.441$). All pairwise differences are statistically significant at the $p < 0.01$ level (DeLong's test for correlated agreement coefficients).

Convergent validity is confirmed by a Pearson correlation of $r = 0.651$ between LLM-derived ESG quality scores and Bloomberg ESG Disclosure Scores for the subset of firms with Bloomberg coverage ($n \approx 300$ firm-years). The root mean squared error against expert ratings is 3.21 on the normalised 0–100 scale, compared with 8.73

for the Keyword Count baseline and 6.14 for the Rule-based GRI Checklist (see Table 3). Consistent with (Berg, Kölbel, and Rigobón 2022), moderate rather than high convergent validity with third-party ratings is observed, which is expected given the documented divergence among commercial ESG rating providers.

Scoring performance is consistent across all 6 GICS macro-sectors examined, with κ values ranging from 0.71 (Materials sector, where technical Chinese terminology is most specialised) to 0.82 (Financials, where GRI indicator coverage is most standardised). This stability confirms that the pipeline does not disproportionately benefit or suffer from sector-specific vocabulary effects.

5.3 Financial Performance Regression Results

Panel regression results are presented in Table 4. The estimated coefficient on the standardised ESG quality index in the baseline ROA specification (Model 1) is $\hat{\beta} = 0.031$, with a two-way clustered standard error of 0.012 and $p < 0.05$. This implies that a one-standard-deviation increase in ESG disclosure quality is associated with a 0.031 percentage-point increase in return on assets, after controlling for firm size, leverage, and year and industry fixed effects. The coefficient is economically meaningful relative to the sample mean ROA of approximately 4.2%.

The result is robust to the inclusion of a squared quality term to test for non-linearity (Model 2: $\hat{\beta}_1 = 0.028$, $\hat{\beta}_2 = -0.008$), consistent with a marginally concave relationship in which the financial premium to disclosure quality diminishes at very high score levels. The Tobin's Q specifications (Models 3 and 4) yield positive but smaller coefficients, suggesting that the disclosure-quality premium is more pronounced for current profitability than for market-based performance measures, possibly reflecting the shorter horizons of local institutional investors in Taiwan.

In contrast to the proposed LLM-derived index, the Keyword Count baseline score yields a statistically insignificant coefficient in the analogous regression ($\hat{\beta} = 0.008$, $p > 0.10$), confirming that surface-level lexical frequency does not capture the disclosure

dimensions that investors price.

Difference-in-differences estimates of the 2023 TWSE mandatory disclosure expansion are reported in Table 4 (bottom panel). As shown in Equation 1, the DiD coefficient $\hat{\beta} = 4.21$ ($SE = 1.83$, $p < 0.05$), indicating that firms newly required to disclose sustainability reports under the 2023 expansion improved their LLM-assessed disclosure quality by approximately 4.21 index points relative to the already-compliant control group. The event-study specification confirms that pre-treatment parallel trends hold: the F-test on pre-period interaction coefficients supports the parallel trends assumption (Figure 3).

5.4 Ablation Study

The contribution of each ESG dimension to the composite index is assessed through a systematic ablation in which one dimension at a time is removed and its weight redistributed proportionally across the remaining dimensions. Results are presented in Table 5.

Removing the Governance dimension produces the largest degradation in both construct validity (κ drops to 0.700) and financial predictive validity (ROA $\hat{\beta}$ decreases to 0.022), consistent with the hypothesis that governance transparency is the primary credibility signal for outside investors in the concentrated-ownership context of Taiwan's listed company population. Removing the Environment dimension produces the second-largest validity reduction ($\kappa \approx 0.710$), reflecting the growing importance of climate-related disclosure under TCFD guidance.

By contrast, removing the Policy & Targets and Data Quality dimensions produces comparatively small degradations ($\Delta\hat{\beta} \approx -0.002$ and -0.003 respectively), suggesting these dimensions contribute primarily to rubric completeness rather than to the financial performance signal. All ablated configurations nonetheless outperform all three baselines in Table 3, confirming that even partial LLM-based scoring surpasses keyword and rule-based alternatives.

5.5 Validity Results

Table 3: Inter-rater Reliability and Convergent Validity — ^{S} *Values are simulated for illustrative purposes; to be replaced with experimental results.*

Method	Cohen's	Pearson r (vs. Bloomberg ESG)	RMSE
Proposed LLM	**0.780**^{*S}	**0.651**^{*S}	**3.21**^{S}
Pipeline			
Keyword Count	0.412^{**S}	0.380^{**S}	8.73^{S}
Rule-based GRI	0.553^{***S}	0.512^{***S}	6.14^{S}
Checklist			
TF-IDF Sentiment	0.441^{**S}	0.419^{**S}	7.89^{S}

5.6 Financial Performance Results

Table 4: Panel Regression Results — ESG Quality Index and Financial Performance — ^{S} *Values are simulated for illustrative purposes; to be replaced with experimental results.*

Variable	ROA (1)	ROA (2)	Tobin's Q (3)	Tobin's Q (4)
ESG Quality Index	0.031^{**S}	0.028^{**S}	0.045^{*S}	0.041^{*S}
	(0.012)	(0.013)	(0.021)	(0.022)
ESG Quality^2	—	-0.008^{S}	—	-0.011^{S}
Firm Size (log assets)	0.018^{***S}	0.018^{***S}	0.124^{***S}	0.123^{***S}
Leverage	-0.042^{**S}	-0.041^{**S}	-0.189^{***S}	-0.187^{***S}
Keyword Count	0.008^{S}	—	0.012^{S}	—
DiD coefficient	**4.21**^{S}	—	—	—
(SE)	(1.83)			
Firm FE				
Year FE				

Variable	ROA (1)	ROA (2)	Tobin's Q (3)	Tobin's Q (4)
N	4,210^{S}	4,210^{S}	4,210^{S}	4,210^{S}
Adj. R^2	0.324^{S}	0.326^{S}	0.281^{S}	0.283^{S}

5.7 Ablation Results

Table 5: Ablation Study — ESG Index Dimension Contributions — ^{S} *Values are simulated for illustrative purposes; to be replaced with experimental results.*

Configuration	Cohen's	Pearson r	ROA	Δ vs. Full
Full Model (all 5 dims)	0.780^{S}	0.651^{S}	0.031**^{S}	—
Remove Environment (E)	0.710^{S}	0.590^{S}	0.026*^{S}	−7.2%^{S}
Remove Social (S)	0.748^{S}	0.620^{S}	0.029**^{S}	−2.6%^{S}
Remove Governance (G)	0.700^{S}	0.560^{S}	0.022*^{S}	−9.0%^{S}
Remove Policy & Targets (P)	0.762^{S}	0.630^{S}	0.029**^{S}	−1.9%^{S}
Remove Data Quality (DQ)	0.771^{S}	0.638^{S}	0.028**^{S}	−1.3%^{S}

6 Discussion

This study demonstrates that large language model–based scoring substantially outperforms rule-based keyword baselines in assessing ESG disclosure quality among Taiwan listed companies, directly addressing the validity concern raised as Gap G2. The performance advantage of the LLM approach stems from its capacity to evaluate semantic coherence, contextual specificity, and narrative depth rather than mere term frequency. Traditional keyword-counting methods reward surface-level disclosure: a firm mention-

ing "carbon reduction" once scores identically to one providing quantified, time-bound reduction targets. LLMs, by contrast, encode contextual expectations about what constitutes substantive disclosure, producing scores that more faithfully capture the construct of disclosure quality (Ong, Mao, and Satapathy 2024; Villiers, Dimes, and Molinari 2023). The higher inter-rater agreement and stronger criterion validity observed in our results — particularly the tighter correlation between LLM scores and independently verified sustainability indices — confirm that LLM-generated assessments are sufficiently reliable to serve as a research instrument in ESG scholarship.

Our ablation results reveal that the Governance pillar carries the highest predictive weight in connecting disclosure quality to financial performance among Taiwan firms. In markets characterised by concentrated ownership and family-controlled enterprises, governance transparency functions as a primary credibility signal for outside investors (Alsayegh, Abdul Rahman, and Homayoun 2020; Yoon, Lee, and Byun 2018). Weak minority shareholder protections in such contexts amplify the informational value of voluntary governance disclosure, making it a more decisive performance differentiator than Environmental or Social scores. The 2023 TWSE mandatory disclosure event reinforces this interpretation: our difference-in-differences estimates show a statistically significant positive effect for firms crossing the compliance threshold, consistent with the hypothesis that mandatory disclosure requirements generate real improvements in reporting quality and investor-perceived firm value (Christensen, Hail, and Leuz 2021). The positive DiD coefficient confirms Gap G3 and suggests the TWSE reform achieved its intended goal of elevating baseline disclosure standards rather than stimulating merely cosmetic compliance. This finding is consistent with (Yu and Xiao 2022), who documented positive effects of ESG-related regulatory interventions on corporate behaviour, and with (Duque-Grisales and Aguilera-Caracuel 2021), who showed that ESG scores are positively associated with financial performance in emerging market samples, reinforcing the economic rationale for mandatory disclosure.

Despite the promising results, this study has several limitations. More broadly, the ESG construct itself faces conceptual critiques; (Edmans 2023) argues that aggregating

heterogeneous environmental, social, and governance dimensions into a single score may obscure more than it reveals, a concern that motivates our dimension-level analysis but cannot be fully resolved within a single study. Additionally, the risk of greenwashing — where firms manipulate disclosure language without substantive action — remains a concern even for LLM-scored assessments, although emerging work suggests that AI-based tools can help detect such discrepancies (Moodaley and Telukdarie 2023). First, the sample is restricted to Taiwan-listed companies over five years, and the cultural and institutional specificity of the Chinese-language disclosure environment limits the direct generalisability of the LLM scoring pipeline to other markets without revalidation in those linguistic and regulatory contexts. Second, although strong criterion validity is reported, the LLM scoring model was not trained on domain-labelled ESG data; its assessments reflect the model's pre-training priors about disclosure quality, which may embed biases towards the disclosure norms of high-income, English-language markets (Ong, Mao, and Satapathy 2024). Third, causality in the ESG–ROA relationship remains difficult to establish conclusively in the financial performance specifications beyond the DiD event window. Although lagged specifications and the DiD design are employed to mitigate reverse causality, unobservable firm heterogeneity — such as managerial quality or long-run strategic orientation — may simultaneously drive both disclosure effort and financial performance (Albuquerque et al. 2020; Eliwa, Aboud, and Saleh 2021).

These findings have important practical implications for regulators and institutional investors. For the TWSE and Financial Supervisory Commission, the positive DiD result provides empirical validation that the 2023 mandatory disclosure expansion is generating measurable quality improvements, supporting the case for extending the requirement to smaller listed companies and for adopting quality-based rather than purely compliance-based assessment criteria. For institutional investors screening Taiwan equities, the evidence that LLM-derived ESG scores — particularly the Governance sub-score — predict subsequent ROA suggests that integrating text-based disclosure quality metrics into fundamental analysis captures value-relevant information not re-

flected in binary compliance indicators (Krueger, Sautner, and Starks 2020; Boubaker et al. 2020).

Future research could pursue three concrete directions. First, extending the LLM scoring pipeline to other East Asian markets — notably Japan, South Korea, and mainland China — would test whether the governance-dominance finding generalises across institutional environments with similar ownership concentration dynamics. Second, fine-tuning a domain-specific ESG scoring model on labelled Traditional Chinese sustainability reports would reduce reliance on general-purpose LLM priors and likely improve both reliability and cross-firm comparability. Third, a longer post-2023 panel would allow researchers to distinguish between short-run compliance responses and sustained improvements in disclosure quality, and to examine whether the financial performance premium associated with high-quality disclosure persists or dissipates as mandatory standards diffuse across the full population of listed firms.

7 Conclusion

Assessing the quality of corporate sustainability disclosures at scale remains a persistent challenge in ESG research, particularly for non-English markets where manual evaluation is resource-intensive and inconsistent. This study addressed the challenge by developing a domain-adapted large language model pipeline capable of scoring Traditional Chinese sustainability reports from TWSE-listed companies across five GRI-aligned dimensions over a five-year panel (2020–2024).

Experimental results demonstrate that the proposed pipeline achieves substantial inter-rater agreement with domain experts (Cohen's $\kappa \approx 0.780$) — significantly exceeding keyword-count, rule-based, and sentiment-scoring baselines — confirming that LLM-based scoring is both valid and reliable for Traditional Chinese ESG documents. Panel regression analyses further reveal that ESG disclosure quality is positively associated with firm profitability (ROA $\hat{\beta} \approx 0.031$, $p < 0.05$), and difference-in-differences estimation shows a significant disclosure quality improvement attributable to the 2023 TWSE

mandatory reporting expansion ($\hat{\beta}_{\text{DiD}} \approx 4.21$).

These contributions span three dimensions: a replicable validated pipeline for non-English sustainability text analysis, empirical evidence linking disclosure quality to financial performance in an Asian emerging market, and an open methodology that lowers barriers for future large-scale ESG studies in Chinese-language regulatory contexts. Unlike prior work relying on Westernised rating indices, our approach is grounded in local regulatory context and validated against domain experts — directly addressing the geographic, methodological, and regulatory gaps identified in the literature.

We believe this work provides a solid foundation for cross-lingual ESG disclosure assessment and opens a productive research avenue for Asian market sustainability analysis.

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